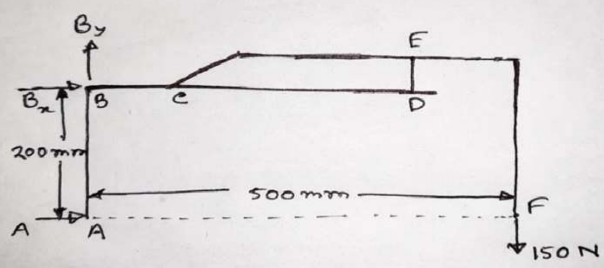
Analysis of Frames

- **Solution:**

FBD of entire frame:



$\circlearrowleft^{+} \Sigma M_B = 0$: $+(A \times 200) - (150 \times 500) = 0$

$\Rightarrow A = 375\ N$

$\therefore \boxed{A = 375\ N \longrightarrow}$

$\xrightarrow{+} \Sigma F_x = 0$: $A + B_x = 0$

$\Rightarrow 375 + B_x = 0$

$\Rightarrow B_x = -375\ N$

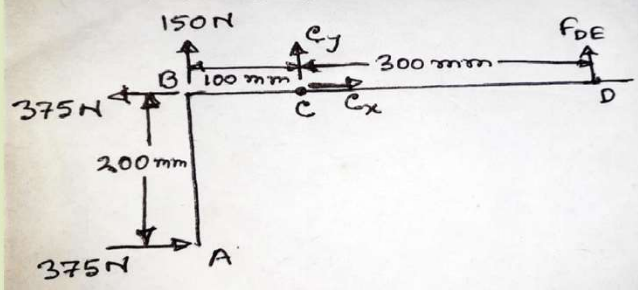
$\therefore \boxed{B_x = 375\ N \longleftarrow}$

$+\uparrow \Sigma F_y = 0$: $B_y - 150 = 0$

$\Rightarrow B_y = 150\ N$

$\therefore \boxed{B_y = 150\ N \uparrow}$

FBD of member ABCD:



$\circlearrowleft^{+} \Sigma M_C = 0$:

$(F_{DE} \times 300) - (150 \times 100) + (375 \times 200) = 0$

$\Rightarrow F_{DE} = -200\ N$

$\therefore \boxed{F_{DE} = 200\ N \downarrow}$

$\xrightarrow{+} \Sigma F_x = 0$:

$C_x + 375 - 375 = 0$

$\therefore \boxed{C_x = 0}$

$+\uparrow \Sigma F_y = 0$: $C_y + 150 + \overset{-200N}{\cancel{F_{DE}}} = 0$

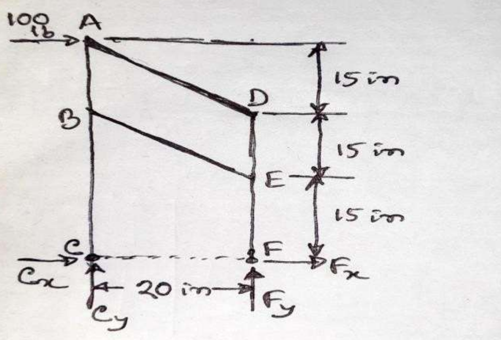
$\Rightarrow C_y = 50\ N$

$\therefore \boxed{C_y = 50\ N \uparrow}$

Analysis of Frames

➢ **Solution:**

FBD of entire frame:



$\circlearrowleft + \Sigma M_C = 0:$ $(F_y \times 20) - (100 \times 45) = 0$

$\Rightarrow F_y = 225 \text{ lb}$

$\therefore \boxed{F_y = 225 \text{ lb} \uparrow}$

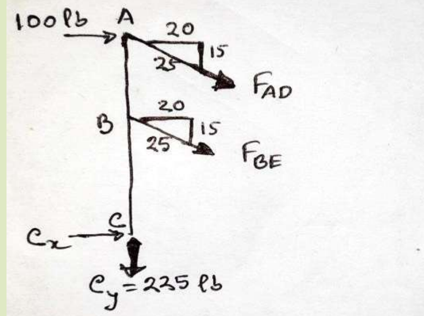
$\circlearrowleft + \Sigma M_F = 0:$ $-(C_y \times 20) - (100 \times 45) = 0$

$\therefore C_y = -225 \text{ lb}$

$\therefore \boxed{C_y = 225 \text{ lb} \downarrow}$

$\xrightarrow{+} \Sigma F_x = 0:$ $C_x + F_x + 100 = 0$ —— (I)

FBD of member 'ABC'



$\xrightarrow{+} \Sigma F_x = 0:$ $C_x + 100 + \frac{20}{25} F_{AD} + \frac{20}{25} F_{BE} = 0$

$\Rightarrow C_x = -100 - \frac{4}{5}(F_{AD} + F_{BE})$ —— (II)

$+\downarrow \Sigma F_y = 0:$ $\overset{225}{\cancel{C_y}} + \frac{15}{25} F_{AD} + \frac{15}{25} F_{BE} = 0$

$\Rightarrow \frac{3}{5}(F_{AD} + F_{BE}) = -225$

$\Rightarrow F_{AD} + F_{BE} = -375$ —— (III)

So, from (II) $\Rightarrow$ $C_x = -100 - \frac{4}{5} \times (-375)$

$\Rightarrow C_x = 200 \text{ lb}$

$\therefore \boxed{C_x = 200 \text{ lb} \rightarrow}$

$\circlearrowleft + \Sigma M_A = 0:$ $(\overset{200}{\cancel{C_x}} \times 45) + (\frac{4}{5} F_{BE} \times 15) = 0$

$\Rightarrow F_{BE} = -750 \text{ lb}$

$\therefore \boxed{F_{BE} = 750 \text{ lb (C)}}$

So, from (III) $\Rightarrow$ $F_{AD} - 750 = -375$

$\Rightarrow F_{AD} = 375 \text{ lb}$

$\therefore \boxed{F_{AD} = 375 \text{ lb (T)}}$

$\therefore C = \sqrt{C_x^2 + C_y^2} = \sqrt{200^2 + 225^2}$

$\therefore \boxed{C = 301.04 \text{ lb}}$

$\theta = \tan^{-1}(C_y / C_x) = \tan^{-1}(\frac{225}{200})$

$\therefore \boxed{\theta = 48.37°}$